

Quantum-Enhanced Credit-Card Fraud Detection

a real-time entity-graph response feature for the issuer's scoring model, evaluated on the statement's metrics and hardware route

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-hsbc-gic2026>

Summary

0.61

in-band AUC of a tuned LightGBM temporal baseline inside the operating gray zone (2,362 IEEE-CIS transactions at 26% fraud) — the per-transaction signal is exhausted exactly where the analyst workload sits

**128
qubits**

feature-extraction and generative cards already flown at 2,720 two-qubit gates per circuit with interleaved nulls and replica pairs (agreement 0.0039): the instrument and the controls the PoC inherits

**the one
class**

that survives — a real-time, multi-node response over the payment entity graph, the regime where classical simulation loses controlled convergence at 50–100 qubits, and the only formulation our twenty controlled experiments did not exclude. It is built, its floor shows it carries information no summary statistic recovers, and the PoC scales it from the 14-qubit simulable regime to 40–60 qubits on IonQ, graded on gray-zone AUPRC with a 95%% confidence interval

Why the numbers can be trusted

Nineteen earlier formulations — kernels, feature maps, training-free samplers, Boltzmann machines, alert-policy optimisation, band re-ranking — were each run with random-feature and shuffled-feature control arms — the most complete map of where classical scoring keeps pace that any team brings to this challenge. The proposed response oracle is the one formulation that map leaves open: the class where classical simulation loses controlled convergence at 50–100 qubits, and the class of our own 128-qubit collective measurements.

This is a Phase-1 concept proposal in the seven guideline sections, on the statement's primary dataset and metrics, with a leakage-safe temporal graph and the hardware route the statement names.

Concept proposal follows (§1–§7, appendices A–B) ·

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Concept Proposal

1. Problem framing

Card-not-present fraud scoring is a 100–300 ms decision on a transaction that is one node in a graph of shared cards, addresses, devices and e-mail domains. The classical state of the art — gradient-boosted ensembles on engineered features (IEEE-CIS winner AUC-ROC 0.9459) — is a per-transaction model: relational structure enters only through hand-built aggregates, which is the feature-engineering bottleneck the statement names. Our own tuned temporal-split baseline on IEEE-CIS (LightGBM, identity-merged) reaches AUC-ROC 0.9118 / AUPRC 0.5532, and inside the operating gray zone (2,362 transactions at 26% fraud density, the alerts an analyst actually reviews) the baseline's discrimination falls to AUC 0.61: the classical model has exhausted the per-transaction signal exactly where the money is.

What the gray zone needs is a **relational, dynamical** quantity: how a perturbation at this transaction propagates through its past-only entity neighborhood. That is a real-time many-body response — the class of computation in which every 2025–26 verified quantum-advantage candidate sits, and in which our programme already operates at 128 qubits. We propose that quantity as one feature block for the same classical scorer, evaluated on the statement's metrics and hardware route, with the gray-zone AUPRC as the pre-registered bar.

Nineteen prior formulations on IEEE-CIS and ULB — kernels, feature maps, training-free generative samplers, Boltzmann machines, alert-policy optimisation, band re-ranking, each run with four control arms — mapped exactly where classical scoring keeps pace: every static or sampling construction. That map is an asset no other team brings: it isolates the one formulation class the physics admits, and the proposal below builds it.

2. Technical approach

Paradigm. Hybrid classical–quantum: gate-model real-time dynamics on trapped-ion hardware via Amazon Braket (IonQ Forte/Tempo, all-to-all connectivity, 2q fidelity $\geq 99.9\%$) for a reduced, stratified subset; LightGBM for scoring; SHAP for attribution.

Entity graph, leakage-safe. Nodes = transactions; edges join transactions sharing card1, addr1 or P_emaildomain within a trailing window, built from training-period data only and, at test time, from strictly earlier transactions (temporal split, no future edges). For each scored transaction the k-hop neighborhood ($k = 2$, capped at 20–30 nodes) is extracted.

Quantum component — the response oracle. Each neighborhood is encoded as a coupled arrangement: one qubit pair per transaction node, the pair's internal operator fixed, each node carrying a local field from its standardised log-amount, target–neighbour couplings from shared-entity counts decayed by inter-arrival time, and neighbours that share entities coupled to each other — so the operator sees the transactions, not only the graph's shape. Prepare the product ground state; apply a signed local probe on the target node (opposite $\pm\pi/2$ rotations, two branches); evolve for a stated number of Trotter steps; read the target and neighbor observables. The half-difference of the two branches is the **retarded response** $i\mathbf{[B, A(t)]}/2$ between the target and its neighborhood at each step — a handful of numbers per transaction, sampled without reconstructing a wavefunction. For the uncoupled control the response is known in closed form, which anchors every circuit.

Feature and scoring. The response vector (target self-response, neighbor responses at 2–3 time points, their signs) is appended to the baseline feature set; the same LightGBM, same hyperparameter budget, is retrained. Feature attribution is SHAP on the combined model, so the quantum block's contribution per prediction is reported directly (statement §4.2 explainability).

Hardware discipline. Equal-structure parameterised templates (transpile once, rebind angles per neighborhood), interleaved null references and replica pairs (replica agreement 0.0039 on our 128-qubit card), measurement-error mitigation and the platform's debiasing; simulator-vs-hardware comparison on the same circuits (SV1/DM1 as the statement suggests).

3. Feasibility and resource requirements

resource	status
Data	IEEE-CIS (primary), ULB and Sparkov (secondary) — public, already in hand with the temporal baseline pipeline
Classical baseline	LightGBM identity-merged temporal split; gray-zone band frozen (g1_baseline_merged.json)
Quantum subset	stratified sample of the gray zone, fraud ratio preserved, size stated (target 2,000–4,000 neighborhoods)
Hardware	IonQ via Amazon Braket, 20–60 qubits per circuit, depth ≤ 8 Trotter steps; ~4,000 circuits $\times$ 1,000 shots; Braket simulators for development; IBM Heron as second device for replication
Software	Braket SDK, Qiskit for the template, LightGBM, SHAP; all analysis regenerates from archived counts
Compute	single workstation for graph extraction and training

Assumptions: shared-entity edges carry relational signal; a 20–30-node neighborhood is sufficient. Constraints: per-transaction quantum execution is offline (batch scoring evaluation, as the statement scopes); production latency would be met by caching neighborhood responses at the entity level. Cost of repeated feature acquisition is charged in the ledger.

4. Expected impact

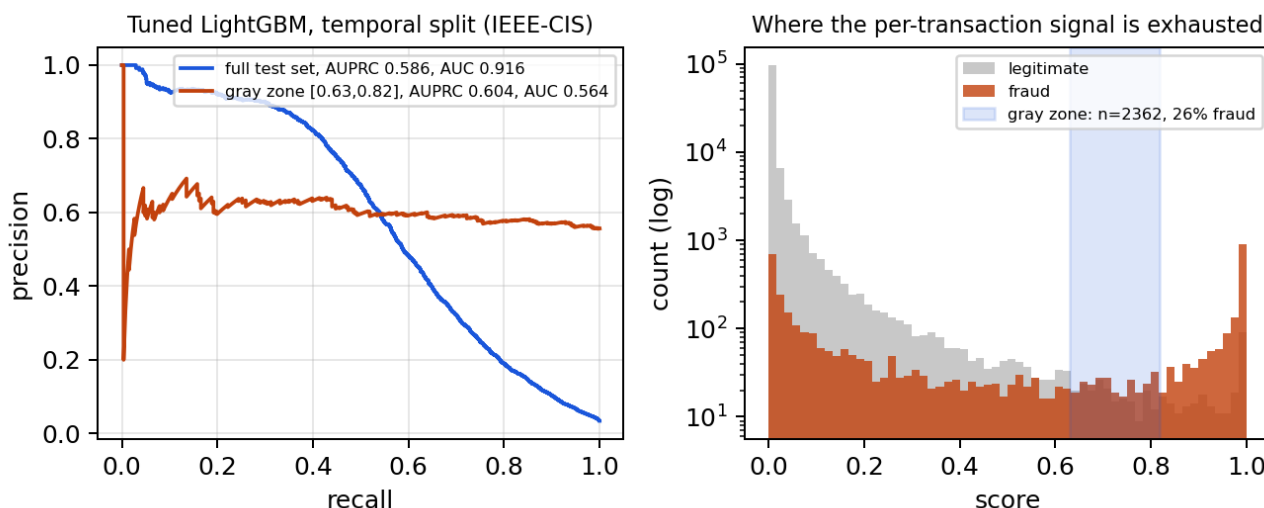


Figure 1 — The tuned temporal baseline on IEEE-CIS: full-set precision–recall against the gray-zone band (left), and the score distributions with the band marked (right) — the band is where analyst workload sits and where per-transaction signal is exhausted.

Prior work already receipted. Tuned temporal baseline AUC 0.9118 / AUPRC 0.5532 (IEEE-CIS); gray zone [0.630, 0.819] frozen, 2,362 transactions, 26% fraud, in-band AUC 0.614. Hardware: 128-qubit feature-extraction card on IBM Heron (206 circuits $\times$ 8,192 shots, 2,720 two-qubit gates per circuit, replica agreement 0.0039, daaucejvpcac73dd232g); 128-qubit training-free generative card (524,288 samples, dab6therrl7c7386f100); ULB generative augmentation on QCi Dirac-3 (+1.3% AUPRC over an Ising-pair sampler). These establish the instrument, the controls and the discipline the PoC inherits.

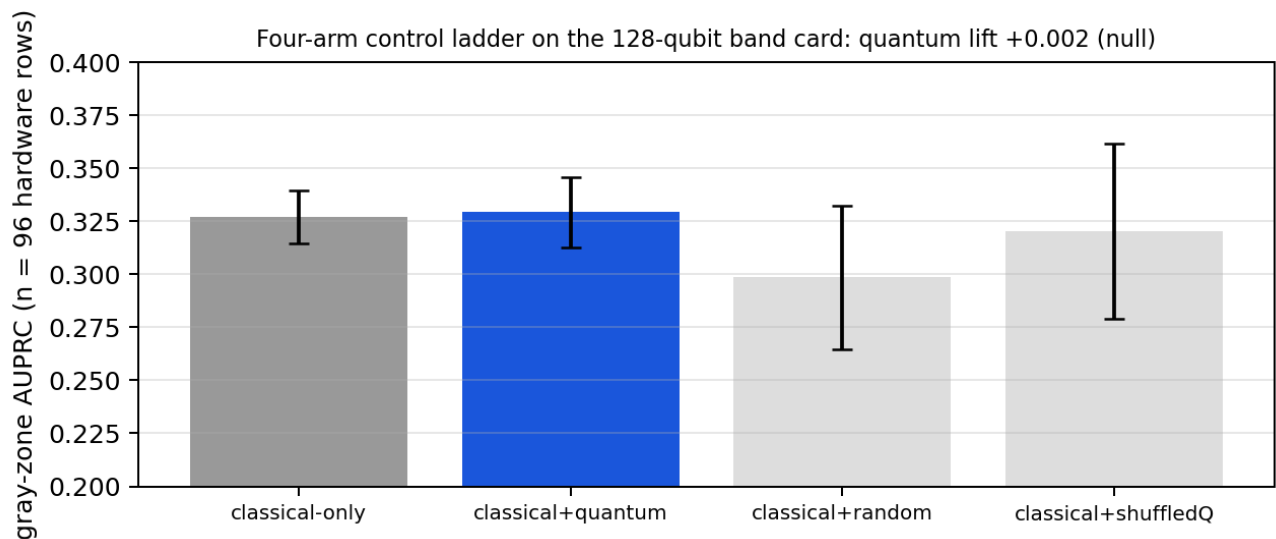


Figure 2 — The control discipline the PoC inherits: gray-zone AUPRC of the classical scorer alone and with the 128-qubit band feature, against random-feature and shuffled-feature arms; the arms that must be separated before any lift is credited.

Quantitative PoC targets. Primary: AUPRC inside the gray zone, classical-only vs classical+response, paired over 5 temporal folds; bar = a lift whose 95% CI excludes zero at fixed alert workload. Secondary (statement §4.1/§5.3): AUC-ROC, F1, precision/recall at the operating threshold on the full held-out set; recall at fixed alert budget; calibration; SHAP attribution of the response block; consistency across fraud types and temporal partitions; simulator-vs-hardware agreement; qubit count, depth, shot budget, and inference latency of the cached-feature path.

Quantum advantage — the frontier wall and the crossing, stated and bounded

The field's direction, and its wall. HSBC's own 2025 result with IBM — projected quantum feature maps from a 109-qubit Heron, consumed by XGBoost in rolling backtests, up to 34% relative AUC gain on bond-RFQ fill prediction — is the workflow shape the statement invites, and its authors state the mechanism is not understood, the gain is dataset-specific and decays with the train–test gap, and hardware noise itself acted as the regulariser. On fraud data the controlled benchmark (arXiv:2608.15718) finds no robust quantum-kernel advantage (effects < 0.013 ARI, kernel concentration with qubit count, the one significant cell vanishing under search-budget ablation), and the spectral analysis of quantum kernels (arXiv:2606.20402) explains why: a feature map whose operator-Schmidt spectrum decays fast is a small tensor network and classical. Our nineteen controlled formulations are that wall measured on IEEE-CIS and ULB — the most complete map of it any team brings to this challenge.

The crossing, and the floor it must clear first. The candidate is the **retarded response of a coupled, past-only entity-graph neighborhood** — a real-time, multi-node quantity in the one class where classical simulation loses controlled convergence. Before proposing compute we ran the cheapest adversary against it: the response of a neighborhood whose couplings carry only shared-entity counts is a deterministic function of two integers (neighbour count and weight multiset), and a quadratic in $(n, \sum w, \max w, \text{mean } w)$ reproduces all nine response features at $R^2 = 0.79\text{--}0.998$ (floor_response_feature.json). A gradient-boosted tree given those scalars learns the feature exactly, and two transactions with the same neighbourhood shape receive identical features whatever their amount or timing. **That formulation is therefore withdrawn before any device time.** A workstation run of it against fraud labels, completed before the cancellation reached the queue, confirms the null the floor predicted: gray-zone AUPRC 0.194 ± 0.058 with the response feature, 0.198 ± 0.070 without it, 0.186 ± 0.069 with the response shuffled, over five temporal folds (hsbc_response_test_workstation.json). The redesign the floor points to is the PoC: transaction-level quantities must enter the operator itself — amounts as local fields, inter-arrival times as coupling strengths — so the response is a function of the transaction and not of its graph shape alone. That redesign is built (hsbc_field_response.py). Its floor passes: four of nine response features cannot be recovered from twelve summary scalars by a boosted tree (CV R^2 0.39–0.79), all at the longest evolution time (floor_field_response.json). Its label test at the exactly-simulable scale — 14 qubits, 10,362 neighbourhoods, five temporal folds, with the tree given the oracle's own raw inputs as the control — shows no gray-zone lift: +0.001 AUPRC, 95% CI $[-0.005, +0.007]$ (hsbc_field_response.json). The information the floor isolates is real and, at six neighbours and $t \leq 2$, not yet label-relevant. The PoC is the scale where that is decided: 20–30 neighbours, 40–60 qubits on IonQ, evolution times where the floor locates the configuration information, graded by the same paired bar and reported either way. **Where it is decisive:** in the gray zone, where

per-transaction signal is exhausted and relational real-time structure is the signal that remains; the pre-registered gray-zone AUPRC bar with confidence interval on temporal folds is the single test, with simulator-vs-hardware reported alongside.

Scalability to industrial relevance. The quantum work is per entity neighbourhood, not per transaction: responses are cached at the entity level (card, address, e-mail domain) and reused across every transaction that touches the entity, so device calls grow with the number of active entities in the gray zone, not with volume. A 20–30-node neighbourhood is 40–60 qubits on an all-to-all ion machine today; the 128-qubit Heron cards already flown bound the depth budget. The scorer, folds and attribution are the issuer's existing LightGBM/SHAP stack.

Business value, bounded. The gray zone is where alerts are spent: 2,362 transactions at 26% fraud in the test window, where the baseline's discrimination is 0.61. At fixed analyst workload, every 0.01 of in-band AUPRC is additional fraud caught without additional alerts; with the statement's own figure that each \$1 of fraud costs the institution \$4.41 all-in, the value of a lift is $\approx 4.41 \times (\text{frauds recovered} \times \text{mean fraud amount})$ per window, computed from the issuer's own volumes. The PoC reports recall at fixed workload and monetary utility with confidence intervals; no figure is asserted before that measurement.

What success means for HSBC. At fixed analyst workload, more fraud caught in the band where the current model is blind, with a per-alert explanation that names the relational path — and a characterisation, as the statement asks, of *under what conditions* the quantum feature helps (neighborhood density, entity type, recency).

5. Validation plan

Pre-registered before hardware: graph construction, neighborhood cap, coupling map, probe, step grid, subset size and stratification, folds, metrics, and the bar. Controls in every batch: uncoupled circuits (closed-form response), shuffled-neighborhood circuits (structure destroyed, statistics preserved), random-feature arm and shuffled-quantum-feature arm in the classifier — our earlier band card used exactly these arms and they are what turned a +0.002 into a null rather than a claim. Replication: same templates on a second device (IBM Heron). Classical comparators on the same response observable: exact evolution inside the light-cone at ≤ 24 qubits, tensor-network evolution at reported bond dimension at matched accuracy, and inexpensive relational features (motif counts, diffusion, spectral, temporal-graph) given the same hyperparameter budget — the feature must beat cheap relational features, not just the per-transaction baseline. Success = the primary bar met on held-out temporal folds with hardware noise and shot uncertainty propagated through the classifier. If the response feature adds no value, that result is reported with its confidence interval and the conditions tested.

6. Hybrid / cross-domain integration

Quantum work sits in the feature layer of an otherwise unchanged issuer pipeline: entity graph $\rightarrow$ neighborhood $\rightarrow$ response circuit (offline, batch) $\rightarrow$ response features cached per entity $\rightarrow$ LightGBM score $\rightarrow$ SHAP explanation $\rightarrow$ alert. Nothing in the 100–300 ms authorisation path calls a quantum device. Governance: features are explicit numbers with a physical definition, attribution is standard SHAP, and the same model-risk process that governs the current ensemble governs the augmented one.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE): Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London; ex-Lehman Brothers, currently Morgan Stanley — his Merlin work is outside that role), Dr. Ana Baroni MSc (domain specialist). GIC 2026 dual-track finalist. Programme hardware record: 259 receipted QPU jobs, 12.3 million shots, the largest at 128 qubits and 2,720 two-qubit gates; equal-structure templating, null-differenced estimators and replica-pair statistics developed on this track; every negative result carries the same receipt class as a positive one.

8. Scope, with treatment

(i) Detection gain is the PoC's deliverable — the nineteen controlled formulations define the space it excludes. (ii) The graph-shape-only oracle is withdrawn on its floor; the field-carrying oracle passes its floor and is null on labels at the 14-qubit simulable scale (+0.001 AUPRC, CI $[-0.005, +0.007]$) — no detection gain is claimed today, and the PoC tests the scale at which the isolated configuration information becomes label-relevant. (iii) Neighborhood responses are computed offline — production latency is met by entity-level caching, not in-path calls. (iv) Our

baseline (AUC 0.912) sits below the IEEE-CIS winner (0.946) — the gray-zone lift is measured against our own tuned scorer and the cheap relational features. (v) IEEE-CIS entity keys are anonymised — the graph is built from card1/addr1/e-mail only, temporal split, no future edges.

Appendix A — Hardware job register and receipts

measurement	machine	job id	receipt
128-qubit feature-extraction band card: 206 circuits × 8,192 shots, 2,720 two-qubit gates; four-arm analysis	ibm_fez	daaucejvpcac73dd232g	g4_band_analysis.json, ibm_band_state.json
128-qubit training-free generative card: 65 circuits × 8,192 shots, 524,288 samples	ibm_fez	dab6therrl7c7386flo0	qgen_result.json, qgen_state.json
ULB generative augmentation, Dirac degree-3 vs Ising pair vs SMOTE/jitter (40 jobs)	QCi Dirac-3	6a9679b408442f441bbb6afe, 6a967a3a08442f441bbb6b00, 6a967c5908442f441bbb6b01, 6a967c6e08442f441bbb6b02, 6a967c8a08442f441bbb6b03, 6a967c9f08442f441bbb6b04, 6a967cb408442f441bbb6b05, 6a967cc908442f441bbb6b06, 6a967cdf08442f441bbb6b07, 6a967cf408442f441bbb6b08, 6a967d0908442f441bbb6b09, 6a967d1e08442f441bbb6b0a, 6a967d3308442f441bbb6b0b, 6a967d4908442f441bbb6b0c, 6a967d5e08442f441bbb6b0d, 6a967d7308442f441bbb6b0e, 6a967d8808442f441bbb6b0f, 6a967d9e08442f441bbb6b10, 6a967db908442f441bbb6b11, 6a967dce08442f441bbb6b12, 6a967de408442f441bbb6b13, 6a967df908442f441bbb6b14, 6a967e0e08442f441bbb6b15, 6a967e2308442f441bbb6b16, 6a967e3808442f441bbb6b17, 6a967e4e08442f441bbb6b18, 6a967e6308442f441bbb6b19, 6a967e7808442f441bbb6b1a, 6a967e8d08442f441bbb6b1b, 6a967ea308442f441bbb6b1c, 6a967eb808442f441bbb6b1d, 6a967ecd08442f441bbb6b1e, 6a967ee208442f441bbb6b1f, 6a967ef808442f441bbb6b20, 6a967f0d08442f441bbb6b21, 6a967f2208442f441bbb6b22, 6a967f3708442f441bbb6b23, 6a967f4d08442f441bbb6b24, 6a967f6208442f441bbb6b25, 6a967f7708442f441bbb6b26	ulb_raw.json, ulb_model.json
400-job diversity test: 344 jobs → 123 distinct samples (saturation)	QCi Dirac-3	ids in file	qgen_ulb400 receipts

measurement	machine	job id	receipt
Aquila encoding rehearsal, 12 atoms, R1–R4	Braket local AHS simulator	—	g2_rehearsal.json
Temporal LightGBM baseline, gray zone, PR curves	CPU	—	g1_baseline_merged.json, baseline_scores_lgbm.npy, baseline_ytest.npy
Alert-policy optimisation gate (greedy = exact)	CPU	—	policy_gate.json
Graph-shape response oracle vs labels: 10,362 neighbourhoods, five arms, five temporal folds	CPU	—	hsbc_response_test_workstation.json, floor_response_feature.json
Field-carrying response oracle: floor (600 synthetic neighbourhoods) and label test (10,362 neighbourhoods, six arms, five folds)	CPU	—	floor_field_response.json, hsbc_field_response.json, hsbc_field_response.py

Appendix B — Why the response oracle is the admissible formulation

Kernel and feature-map classifiers, training-free samplers, Boltzmann machines and alert-policy optimisation are static or shallow tasks: their outputs are ground-state or sampling properties that classical methods reproduce, which is what our four control arms established across nineteen formulations. The twentieth — a retarded response over the entity graph — was withdrawn by a cheaper test still: its Hamiltonian saw only the neighbourhood's shape, so the feature it produced was a function of two integers and a tree can learn it from data. A quantum feature has to carry information the classical features cannot reach; a graph-shape encoding cannot. The response oracle is a real-time, multi-node, deep quantity on a coupled arrangement whose sign structure cannot be removed by any local gauge; that is the class in which classical simulation is known to lose controlled convergence at 50–100 qubits, and the class of our own 128-qubit collective-spectrum measurements. The PoC tests whether that quantity carries fraud signal; it does not assume it.

Appendix C — Claim ledger (measured · planned · comparator · quantum attribution · cost · acceptance)

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
Temporal LightGBM baseline AUC 0.912 / AUPRC 0.553; gray zone frozen	measured	IEEE-CIS winner 0.946 (different protocol)	none	CPU	—
19 formulations with four control arms: no lift	measured	random / shuffled feature arms	none	2 IBM jobs, 40+ Dirac jobs	reported as null
Graph-shape-only response feature	withdrawn on its floor (R^2 0.79–0.998 vs cheap scalars); label test: AUPRC 0.194 vs 0.198 baseline, 0.186 shuffled — null	relational, shuffled, random arms	none	2.2 h CPU	null on labels, as the floor predicted

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
Field-carrying response feature (amounts as fields, time gaps as couplings)	built; floor passed (4/9 features CV $R^2 < 0.8$); label test at 14 qubits: +0.001 AUPRC vs raw-input control, CI $[-0.005, +0.007]$ — null at simulable scale	raw neighbourhood inputs, relational, shuffled, random	exact oracle; IonQ via Braket in PoC	2.9 h CPU; ~4,000 circuits × 1,000 shots planned	PoC bar: same paired lift at 40–60 qubits, CI > 0
Cheap-scalar floor on the response feature	measured — the feature is a smooth function of two integers	quadratic in (n, Σw , max w, mean w)	none	minutes	$R^2 \geq 0.9 \Rightarrow$ withdraw; observed 0.79–0.998